

AB POWER ENGINEERING

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HUMAN POWER

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1. The rate of work done by human body.

- a. Human work
- ☒ b. Human Power
- c. Human Energy
- d. None of the above

2. The amount of power available in human for useful work.

- a. 0.5 hp
- ☒ b. 0.25 hp
- c. 0.1 hp
- d. None of the above

3. If the average power available in human is 0.1 hp, how much power a 16-year old person can generate?

- a. 0.20 5 hp
- ☒ b. 0.152 hp
- c. 0.115 hp
- d. None of the above

4. The amount of power human can develop from the food he eats.

- a. 0.12 hp
- ☒ b. 0.25 hp
- c. 0.50 hp
- d. None of the above

5. The minimum energy requirement of human.

- a. 1,250 cal/day
- b. 1,500 cal/day
- ☒ c. 1,750 cal/day
- d. 2,000 cal/day
- e. None of the above

6. A person is doing a work for 4 hours, what is the power developed by that person?

- a. 0.09 hp
- ☒ b. 0.13 hp
- c. 0.15 hp
- d. None of the above

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7. How much power can a human generate for 8-hour working period?

- ☒ a. 0.095 hp
- b. 0.103 hp
- c. 0.152 hp
- d. None of the above

8. A ten-hectare farm is to be planted with rice by broadcasting. If seeding is required to be finished within one day, how many people would you recommend to do the job?

- ☒ a. 3 persons
- b. 5 persons
- c. 6 persons
- d. None of the above

9. Manpower requirement in transplanting rice.

- a. 15-20 man-days / hectare
- ☒ b. 20-25 man-days / hectare
- c. 25-30 man-days / hectare
- d. None of the above

10. The conversion efficiency of human.

- a. 10%
- b. 15%
- ☒ c. 20%
- d. 25%
- e. None of the above

11. The power human can develop by pedaling.

- ☒ a. 0.27 to 0.53 hp
- b. 0.54 to 0.75 hp
- c. 0.76 to 1.00 hp
- d. None of the above

12. Which of the following is not a manually-operated machine?

- a. Mortar and pestle
- b. Pedal thresher
- c. Dibbler
- ☒ d. None of the above

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13. Human efficiency during hot and during humid condition ____.

- a. increases
- ☒ b. decreases
- c. remains the same
- d. None of the above

14. Ten hectares of rice farm is to be harvested manually in 2 days using sickles (8hr/day). If the field efficiency is 80% and the man-hour/hectare for harvesting is 60, how many persons will be needed to harvest the rice from the 10-hectare farm?

- a. 27 persons
- b. 37 persons
- ☒ c. 47 persons
- d. None of the above

$$\frac{10 \times 80}{(2 \times 8)(0.8)}$$

15. The amount of food energy needed to balance energy expenditure in order to maintain body size, body composition and a level of necessary and desirable physical activity consistent with long-term good health.

- a. Food intake
- b. Energy balance
- ☒ c. Energy requirement
- d. None of the above

16. The average energy spent in a 24-hour period by an individual or groups of individuals.

- ☒ a. Total energy expenditures
- b. Average energy expenditure
- c. Human energy expenditure
- d. None of the above

17. The major advantage of human compared with other sources of power.

- ☒ a. Intelligence
- b. More power available
- c. Longer working time
- d. None of the above

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18. The disadvantage of human power compared with animals and machines is ____.

- a. intelligence
- ☒ b. low power available
- c. All of the above
- d. None of the above

19. Components of energy requirement of human.

- a. Metabolism
- b. Metabolic response to food
- c. Physical activity
- d. Growth
- ☒ e. All of the above
- f. None of the above

20. The indicator of weight adequacy in relation to height of human.

- ☒ a. Body mass index
- b. Body weight-to-height ratio
- c. Body adequacy ratio
- d. None of the above

21. A method to measure the daily energy expenditure of human based on the relationship of heart rate and oxygen consumption and on minute-by-minute monitoring of heart rate.

- a. Heart and oxygen monitoring
- b. Daily energy expenditure monitoring
- ☒ c. Heart rate monitoring
- d. None of the above

22. A human-power implement without moving parts that facilitates mechanical manual operation.

- ☒ a. Tool
- b. Machine
- c. Instrument
- d. None of the above

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